

Qualification Pack



Agrivoltaics Installer Electrical

QP Code: SGJ/Q0122

Version: 1.0

NSQF Level: 4

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Qualification Pack

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SGJ/Q0122: Agrivoltaics Installer Electrical

Brief Job Description

Agrivoltaics Installer Electrical is responsible for installation, operation and maintenance of solar photovoltaic systems for producing power and suitable crops at the project site.

Personal Attributes

This Qualification tells us about combine benefits of solar pv system along with agriculture.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N0164: Perform Site Survey for installation of Solar PV System for Agrovoltaic system](#)
2. [SGJ/N0165: Procure Solar PV system components according to Agrovoltaic Installation](#)
3. [SGJ/N0166: Introduce tools and tackles used in Solar PV installation for Agrivoltaic system installation](#)
4. [SGJ/N0167: Install Electrical components of Solar PV System for Agrivoltaic system installation](#)
5. [SGJ/N0168: Test and Commission Solar PV System and inspect Agrivoltaic System.](#)
6. [SGJ/N0106: Maintain Personal Health & Safety at project site](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Renewable Energy
Country	India
NSQF Level	4
Credits	17

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7421.1401
Minimum Educational Qualification & Experience	12th grade Pass (or equivalent) with NA of experience OR Pursuing 3rd year of 3-year diploma after 10th ((Electrical/ Mechanical)) with NA of experience OR 10th grade pass OR Previous relevant Qualification of NSQF Level (3.5) OR Previous relevant Qualification of NSQF Level (3)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-04-ES-02029-2024-V1-SCGJ
NQR Version	1

Remarks:

Total: 510 Notional hours (Theory : 250 + Practical : 170 and 90 hours of OJT)

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SGJ/N0164: Perform Site Survey for installation of Solar PV System for Agrovoltaic system

Description

How to assess the site condition for the installation of Agri PV system. Identify load to be connected to agrivoltaic System. Perform all pre-site testing activities.

Scope

The scope covers the following :

- This unit tells about Site Survey for installation of Solar PV System for Agrovoltaic system

Elements and Performance Criteria

Site Survey for the installation of agrivoltaic system and access the customer's requirement.

To be competent, the user/individual on the job must be able to:

- PC1.** understand the concept and requirements of agrivoltaic system.
- PC2.** discuss how to Understand the location of installations.
- PC3.** explain how to Select a shadow-free location for agrivoltaic system.
- PC4.** assess the site level pre-requisites for solar PV installation and show how to prepare a site map of the location where installation and cultivation has to be carried out.
- PC5.** identify and list the load to be connected to the Solar PV system and show how to assess the load to be run on agrivoltaic system.
- PC6.** explain how to Check for any shading obstacles to the solar panels.
- PC7.** discuss about ON Grid and Off Grid Solar PV system
- PC8.** explain and perform Soil test, Earth Resistance test, Pull-out test and contoured Survey.
- PC9.** decide on the type of mounting and spacing of panels to be developed.
- PC10.** discuss how to inform the customer of any civil construction to be undertaken for installing the panels.
- PC11.** prepare a load profile.
- PC12.** Outline the importance of Agri PV system.
- PC13.** Perform Soil test, Earth Resistance test, Pull-out test and contoured Survey.
- PC14.** show how to document the site survey variables and complete the checklist/site survey form.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Company documentation policy.
- KU2.** Document information using appropriate corporate forms.
- KU3.** Obtain authorization from specified field safety officer and supervisor
- KU4.** Company reporting structure

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KU5. Organization culture.

KU6. Definition of the terms: energy and power, cell, module, string, array, monocrystalline, polycrystalline, amorphous silicon.

KU7. legislative, organizational, site requirements and procedures

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. complete documentation work applicable to the role

GS2. read various color codes, as per standard electrical, mechanical and civil nomenclature

GS3. follow organisational code of conduct

GS4. make timely decisions for efficient utilization of resources

GS5. • Express statements or information clearly so that others can hear and
• understand.

GS6. Participate in and understand the main points of simple discussions.

GS7. Respond appropriately to any queries.

GS8. Recognize problems and search for solutions.

GS9. Ask questions for better understanding.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Site Survey for the installation of agrivoltaic system and access the customer's requirement.</i>	30	20	-	-
PC1. understand the concept and requirements of agrivoltaic system.	2	-	-	-
PC2. discuss how to Understand the location of installations.	4	-	-	-
PC3. explain how to Select a shadow-free location for agrivoltaic system.	3	-	-	-
PC4. assess the site level pre-requisites for solar PV installation and show how to prepare a site map of the location where installation and cultivation has to be carried out.	3	4	-	-
PC5. identify and list the load to be connected to the Solar PV system and show how to assess the load to be run on agrivoltaic system.	3	4	-	-
PC6. explain how to Check for any shading obstacles to the solar panels.	3	-	-	-
PC7. discuss about ON Grid and Off Grid Solar PV system	3	-	-	-
PC8. explain and perform Soil test, Earth Resistance test, Pull-out test and contoured Survey.	3	-	-	-
PC9. decide on the type of mounting and spacing of panels to be developed.	3	-	-	-
PC10. discuss how to inform the customer of any civil construction to be undertaken for installing the panels.	3	-	-	-
PC11. prepare a load profile.	-	3	-	-
PC12. Outline the importance of Agri PV system.	-	3	-	-
PC13. Perform Soil test, Earth Resistance test, Pull-out test and contoured Survey.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. show how to document the site survey variables and complete the checklist/site survey form.	-	3	-	-
NOS Total	30	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0164
NOS Name	Perform Site Survey for installation of Solar PV System for Agrovoltaic system
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N0165: Procure Solar PV system components according to Agrovoltaic Installation

Description

How to estimate Solar PV system sizing. Final designing and drawing approval. How to Prepare Bill of Material. Discuss Order placing for components, procurement, and verification.

Scope

The scope covers the following :

- This unit tells about how to Procure Solar PV system components according to Agrovoltaic Installation

Elements and Performance Criteria

Procure the agrivoltaic system components and prepare bill of material

To be competent, the user/individual on the job must be able to:

- PC1.** determine size of solar panel, Controller/Inverter, Transformer, Cable, Breaker, SCADA system, etc.
- PC2.** discuss procedure of Design finalization and drawings release
- PC3.** prepare Bill of Materials (BoM) from single line diagram, civil/mechanical and electrical/agriculture drawings and show how to check materials received as per final BOM and ensure that the correct material for the job arrives on site and is damage free
- PC4.** explain how to Approach stores of the company or the market to place the requirement for components as per BOM.
- PC5.** ensure that the quantity of panels, controller/inverter and batteries match the voltage requirement of the system.
- PC6.** document variation and submit to design team (if required) and obtain approval or revised drawings.
- PC7.** ensure that only company recommended quality materials are used unless specified by customer.
- PC8.** ensure that all materials are Quality control passed.
- PC9.** ensure that all the components are handled and stored properly as per standard operating procedures and show how to report and document the status of material received at site and take appropriate action for replacements, if any.
- PC10.** show the connection practice of on-grid/off-grid/hybrid system.
- PC11.** show how to identify and verify the equipment specifications.
- PC12.** show how to arrange for tools and consumables required for mounting the solar panels over crops.
- PC13.** show how to complete all documentation w.r.t. procurement

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Knowhow of Tools & Tackles required for installation.
- KU2.** DOs and Donts of material handling and storage.
- KU3.** legislative, organizational, site requirements and related procedures
- KU4.** Document information using appropriate corporate forms.
- KU5.** Company work safety policy
- KU6.** Companys documentation policy
- KU7.** Company reporting structure and Organization culture

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** undertake complete documentation
- GS2.**
 - correctly read and interpret various color codes, as per standard electrical, mechanical and
 - civil nomenclature
- GS3.** follow organizational rule-based decision making process
- GS4.** approach relevant authority when required
- GS5.** work constructively and collaboratively with others
- GS6.** express statements with required clarity
- GS7.** make timely decisions for efficient utilization of resources

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Procure the agrivoltaic system components and prepare bill of material</i>	23	27	-	-
PC1. determine size of solar panel, Controller/Inverter, Transformer, Cable, Breaker, SCADA system, etc.	4	-	-	-
PC2. discuss procedure of Design finalization and drawings release	4	-	-	-
PC3. prepare Bill of Materials (BoM) from single line diagram, civil/mechanical and electrical/agriculture drawings and show how to check materials received as per final BOM and ensure that the correct material for the job arrives on site and is damage free	2	5	-	-
PC4. explain how to Approach stores of the company or the market to place the requirement for components as per BOM.	2	-	-	-
PC5. ensure that the quantity of panels, controller/inverter and batteries match the voltage requirement of the system.	2	-	-	-
PC6. document variation and submit to design team (if required) and obtain approval or revised drawings.	2	-	-	-
PC7. ensure that only company recommended quality materials are used unless specified by customer.	2	-	-	-
PC8. ensure that all materials are Quality control passed.	2	-	-	-
PC9. ensure that all the components are handled and stored properly as per standard operating procedures and show how to report and document the status of material received at site and take appropriate action for replacements, if any.	3	5	-	-
PC10. show the connection practice of on-grid/off-grid/hybrid system.	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. show how to identify and verify the equipment specifications.	-	4	-	-
PC12. show how to arrange for tools and consumables required for mounting the solar panels over crops.	-	4	-	-
PC13. show how to complete all documentation w.r.t. procurement	-	4	-	-
NOS Total	23	27	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0165
NOS Name	Procure Solar PV system components according to Agrovoltaic Installation
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N0166: Introduce tools and tackles used in Solar PV installation for Agrivoltaic system installation

Description

This unit tells about tools and tackles used in Solar PV installation for Agrivoltaic system Installation

Scope

The scope covers the following :

- Identify tools used in system installation
- Various application of tools and show how to utilise tools for Agro PV system Installation

Elements and Performance Criteria

Introduce tools and tackles used in Solar PV installation for Agrivoltaic system installation

To be competent, the user/individual on the job must be able to:

- PC1.** discuss and demonstrate the role and requirement of tools in solar PV installation.
- PC2.** discuss the various tools types and demonstrate the working of tools.
- PC3.** discuss and demonstrate the specific function, working and application of various tools and the working of tools
- PC4.** describe the tools used for Electric wiring, Solar Panel Installation, load analysis.
- PC5.** discuss about checking and detection of damaged Tools and demonstrate the risk associated with tools.
- PC6.** discuss about all measuring instruments, multimeter, megger, clampmeter, etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Know-how of Tools & Tackles required for Solar PV installation for Agrovoltaic system installation
- KU2.** legislative, organizational, site requirements and related procedures
- KU3.** usage and handling procedure of system components
- KU4.** different mechanical and electrical equipment, their functioning and precautions
- KU5.** tools involved in installation and O&M activities
- KU6.** cleaning procedures for various equipment
- KU7.** Obtain authorization from specified field safety officer and supervisor
- KU8.** Document information using appropriate corporate forms

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** Fill up documentation applicable to one's role.
- GS2.** Read English and/or vernacular language.
- GS3.** Understand the various color codes, as per standard electrical,
- GS4.**
 - Express statements or information clearly so that others can hear and
 - understand.
- GS5.** Respond appropriately to any queries.
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** follow organisational code of conduct

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduce tools and tackles used in Solar PV installation for Agrivoltaic system installation</i>	28	22	-	-
PC1. discuss and demonstrate the role and requirement of tools in solar PV installation.	4	5	-	-
PC2. discuss the various tools types and demonstrate the working of tools.	4	5	-	-
PC3. discuss and demonstrate the specific function, working and application of various tools and the working of tools	5	5	-	-
PC4. describe the tools used for Electric wiring, Solar Panel Installation, load analysis.	5	-	-	-
PC5. discuss about checking and detection of damaged Tools and demonstrate the risk associated with tools.	5	7	-	-
PC6. discuss about all measuring instruments, multimeter, megger, clampmeter, etc.	5	-	-	-
NOS Total	28	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0166
NOS Name	Introduce tools and tackles used in Solar PV installation for Agrivoltaic system installation
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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SGJ/N0167: Install Electrical components of Solar PV System for Agrivoltaic system installation

Description

This unit tells about how to Install Electrical components of Solar PV System for Agrivoltaic system installation

Scope

The scope covers the following :

- Activities required for Agrovoltaic system installation
- Show how to perform Installation of electrical components along with conduits and cables.

Elements and Performance Criteria

Install Electrical components of Solar PV System for Agrivoltaic system installation

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to use the site safety plan and maintain clean work place.
- PC2.** explain how to clarify the maximum working voltage.
- PC3.** explain how to measure current and voltage on equipment before proceeding with work.
- PC4.** inspect and demonstrate the use of electrical installation toolkit.
- PC5.** inspect and maintain safety equipment.
- PC6.** inspect and maintain testing equipment.
- PC7.** inspect the civil and mechanical installation for agrivoltaic system.
- PC8.** install solar panels on the mounting structure with appropriate configuration and show how to install mounting structure on foundation.
- PC9.** maintain proper space between the rows for Solar PV installation and show how to install multiple panels on structure.
- PC10.** select the location for the energy meter, DC combiner box, controller/inverter, battery bank, and earthing point as per the drawing and show how to install panels, controller/inverter and other components.
- PC11.** confirm battery bank location and Install batteries (if required) and show how to confirm and install battery bank racks.
- PC12.** prepare battery terminals and Install battery interconnection cables and show how to Install batteries and prepare battery terminals (e.g., clean).
- PC13.** install DC combiner box and show how to connect cables in the agrivoltaic system.
- PC14.** Prepare conduit and cable routing plan and show how to label and sealing of all the components and cables.
- PC15.** install controller/inverter (as per requirement)
- PC16.** install all cables in the agrivoltaic system as per approved drawing and Show how to select the correct cable type, color, and gauge. Show how to support and secure conduit. Show how to terminate cables.

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- PC17.** test final assembled of panels, controller/inverter, battery, and transformer.
- PC18.** discuss how to install AC combiner box (if required)
- PC19.** discuss how to install utility required disconnects
- PC20.** discuss how to connect all accessories with an earthing system at least more than 2 points.
- PC21.** get the grounding system installed for panels/mounting system, and controller/inverters and show how to check cables for continuity. Show how to locate underground hazards, if any. Show how to determine grounding conductor size.
- PC22.** get the Bonding done for all electrical equipment's and apply anti - oxidant material.
- PC23.** show how to terminate fine stranded cables.
- PC24.** Prepare conduit and cable routing plan and Show how to label and sealing of all the components and cables.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Mechanical and electrical features necessary for the long life of the PV
• system under a wide range of operating conditions
- KU2.** • Determine the type of mounting structure required depending on the
• type of site conditions
- KU3.** Determining the cabling route and estimate the length of cable required.
- KU4.** • Determining where the array junction box (if required) and inverter will
• be located.
- KU5.** • Do's and don'ts of DC wiring and installation of other electrical
• components
- KU6.** • Installation work on a PV power system in accordance with relevant
• standards and regulations
- KU7.** legislative, organizational, site requirements and related technical procedures
- KU8.** obtain relevant authorization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Fill up documentation applicable to one's role.
- GS2.** Read English and/or vernacular language.
- GS3.** Understand the various color codes, as per standard electrical,
- GS4.** express statements with required clarity
- GS5.** make timely decisions for efficient utilization of resources
- GS6.** follow organisational code of conduct

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Install Electrical components of Solar PV System for Agrivoltaic system installation</i>	65	35	-	-
PC1. discuss how to use the site safety plan and maintain clean work place.	3	-	-	-
PC2. explain how to clarify the maximum working voltage.	3	-	-	-
PC3. explain how to measure current and voltage on equipment before proceeding with work.	3	-	-	-
PC4. inspect and demonstrate the use of electrical installation toolkit.	3	3	-	-
PC5. inspect and maintain safety equipment.	3	-	-	-
PC6. inspect and maintain testing equipment.	3	-	-	-
PC7. inspect the civil and mechanical installation for agrivoltaic system.	3	-	-	-
PC8. install solar panels on the mounting structure with appropriate configuration and show how to install mounting structure on foundation.	3	3	-	-
PC9. maintain proper space between the rows for Solar PV installation and show how to install multiple panels on structure.	3	3	-	-
PC10. select the location for the energy meter, DC combiner box, controller/inverter, battery bank, and earthing point as per the drawing and show how to install panels, controller/inverter and other components.	3	3	-	-
PC11. confirm battery bank location and Install batteries (if required) and show how to confirm and install battery bank racks.	3	3	-	-
PC12. prepare battery terminals and Install battery interconnection cables and show how to Install batteries and prepare battery terminals (e.g., clean).	3	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. install DC combiner box and show how to connect cables in the agrivoltaic system.	3	3	-	-
PC14. Prepare conduit and cable routing plan and show how to label and sealing of all the components and cables.	3	3	-	-
PC15. install controller/inverter (as per requirement)	3	-	-	-
PC16. install all cables in the agrivoltaic system as per approved drawing and Show how to select the correct cable type, color, and gauge. Show how to support and secure conduit. Show how to terminate cables.	3	3	-	-
PC17. test final assembled of panels, controller/inverter, battery, and transformer.	3	-	-	-
PC18. discuss how to install AC combiner box (if required)	3	-	-	-
PC19. discuss how to install utility required disconnects	2	-	-	-
PC20. discuss how to connect all accessories with an earthing system at least more than 2 points.	3	-	-	-
PC21. get the grounding system installed for panels/mounting system, and controller/inverters and show how to check cables for continuity. Show how to locate underground hazards, if any. Show how to determine grounding conductor size.	2	3	-	-
PC22. get the Bonding done for all electrical equipment's and apply anti - oxidant material.	2	-	-	-
PC23. show how to terminate fine stranded cables.	-	2	-	-
PC24. Prepare conduit and cable routing plan and Show how to label and sealing of all the components and cables.	2	3	-	-
NOS Total	65	35	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0167
NOS Name	Install Electrical components of Solar PV System for Agrivoltaic system installation
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N0168: Test and Commission Solar PV System and inspect Agrivoltaic System.

Description

This unit tells about how to Test and Commission Solar PV System and inspect Agrivoltaic System.

Scope

The scope covers the following :

- show how to test the Agrovoltaic system.
- how to perform final check before commissioning of Agrovoltaic system
- How to perform the Commissioning of Agrovoltaic system.

Elements and Performance Criteria

Test and Commission Solar PV System and inspect Agrovoltaic System

To be competent, the user/individual on the job must be able to:

- PC1.** discuss how to perform visual inspection of the installed components.
- PC2.** discuss how to inspect civil, mechanical and electrical installation at Agrivoltaic site.
- PC3.** verify the configuration of the panels for crop development.
- PC4.** verify system grounding and measure insulation resistance.
- PC5.** discuss how to check continuity of the system and verify polarity.
- PC6.** discuss how to measure DC voltages and currents for each string and array for proper operation of the system.
- PC7.** verify inverter operation including anti-islanding performance and measure AC system values.
- PC8.** verify calibration of data acquisition system.
- PC9.** preparation of the inspection report of Agrivoltaic system and take appropriate action.
- PC10.** show how to check all disconnects are in (OFF) position during the final checkout.
- PC11.** show how to check terminals connections and screws are securely tightened.
- PC12.** show how to verify installations are matched to the manufacturer's specifications and recommendations.
- PC13.** show how to check applicable warning signs and labels are posted appropriately.
- PC14.** show how to verify the job site is clean and orderly.
- PC15.** show how to connect power source to system., all disconnect or and fuse must be open.
- PC16.** Show how to test DC Voltage and Polarity.
- PC17.** Show how to Test AC voltage at inverter end and Show how to Test Line to Line and Line to Neutral.
- PC18.** Show how to Lift AC disconnect lever (inverter to grid) and Show how to Lift DC disconnect lever (PV to inverter)
- PC19.** Show how to Check normal operation status of each components.

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PC20. Show how to find fault in the system at site.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Typical faults, their causes and resolution for all system components
- KU2.**
 - visual inspection, continuity of wiring, Earthing, polarity check, insulation and
 - voltage drop
- KU3.**
 - Do's and don'ts of DC wiring and installation of other electrical
 - components
- KU4.**
 - Installation work on a PV power system in accordance with relevant
 - standards and regulations
- KU5.** Know-how of Tools & Tackles required for Test and Commission Solar PV System
- KU6.** Diagnostic/fault finding techniques.
- KU7.** Legislative, organization, site requirements and procedures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Fill up documentation applicable to one's role.
- GS2.** Read English and/or vernacular language
- GS3.** Understand the various color codes, as per standard electrical
- GS4.** Participate in and understand the main points of simple discussions
- GS5.** Respond appropriately to any queries.
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** follow organisational code of conduct

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Test and Commission Solar PV System and inspect Agrovoltaic System</i>	18	32	-	-
PC1. discuss how to perform visual inspection of the installed components.	2	-	-	-
PC2. discuss how to inspect civil, mechanical and electrical installation at Agrivoltaic site.	2	-	-	-
PC3. verify the configuration of the panels for crop development.	2	-	-	-
PC4. verify system grounding and measure insulation resistance.	2	-	-	-
PC5. discuss how to check continuity of the system and verify polarity.	2	-	-	-
PC6. discuss how to measure DC voltages and currents for each string and array for proper operation of the system.	2	-	-	-
PC7. verify inverter operation including anti-islanding performance and measure AC system values.	2	-	-	-
PC8. verify calibration of data acquisition system.	2	-	-	-
PC9. preparation of the inspection report of Agrivoltaic system and take appropriate action.	2	-	-	-
PC10. show how to check all disconnects are in (OFF) position during the final checkout.	-	3	-	-
PC11. show how to check terminals connections and screws are securely tightened.	-	3	-	-
PC12. show how to verify installations are matched to the manufacturer's specifications and recommendations.	-	3	-	-
PC13. show how to check applicable warning signs and labels are posted appropriately.	-	3	-	-
PC14. show how to verify the job site is clean and orderly.	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. show how to connect power source to system., all disconnect or and fuse must be open.	-	3	-	-
PC16. Show how to test DC Voltage and Polarity.	-	2	-	-
PC17. Show how to Test AC voltage at inverter end and Show how to Test Line to Line and Line to Neutral.	-	3	-	-
PC18. Show how to Lift AC disconnect lever (inverter to grid) and Show how to Lift DC disconnect lever (PV to inverter)	-	3	-	-
PC19. Show how to Check normal operation status of each components.	-	3	-	-
PC20. Show how to find fault in the system at site.	-	3	-	-
NOS Total	18	32	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0168
NOS Name	Test and Commission Solar PV System and inspect Agrivoltaic System.
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

SGJ/N0106: Maintain Personal Health & Safety at project site

Description

This unit is about maintaining health, safety and hygiene at workplace.

Scope

The scope covers the following :

- Adopt safe practices at workplace
- Follow emergencies, rescue and first aid procedures
- Follow good housekeeping practices and infection control guidelines

Elements and Performance Criteria

Adopt safe practices at workplace

To be competent, the user/individual on the job must be able to:

- PC1.** explain the requirements for safe work area
- PC2.** identify and report any hazards, risks or breaches in site safety to the appropriate authority
- PC3.** follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk
- PC4.** use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition
- PC5.** follow safe practices when working at height and in confined space
- PC6.** handle all required tools, tackles, materials and equipment safely
- PC7.** identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures
- PC8.** apply ergonomic principles wherever required
- PC9.** use safety signs, labels, charts and notices at workplace
- PC10.** identify work safety procedures and instructions for handling heavy components

Follow emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow emergency and evacuation procedures in case of accidents, fires and natural calamities
- PC12.** use appropriate fire extinguishers for different types of fire
- PC13.** administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.
- PC14.** use correct method to move injured person during an emergency

Follow good housekeeping practices and infection control guidelines

To be competent, the user/individual on the job must be able to:

- PC15.** follow recommended personal hygiene, workplace hygiene and sanitation practices
- PC16.** clean and disinfect all material, tools and supplies before and after use

Qualification Pack

- PC17.** report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues
- PC18.** follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of safety drills
- KU2.** importance of working in clean and safe environment
- KU3.** health and safety roles and responsibilities of relevant personnel within and outside the organization
- KU4.** reporting procedures in case of breaches or hazards for site safety, accidents and emergency situations
- KU5.** basic ergonomic principle
- KU6.** key internal and external source of health and safety information
- KU7.** meaning of hazards, risk and near miss
- KU8.** importance of Personal Protective Equipment required for specific job
- KU9.** forms and classification of hazardous substances
- KU10.** health effect associated with exposure to environmental pollution
- KU11.** housekeeping activities relevant to task
- KU12.** symptoms of infection like fever, cough, swelling and inflammation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data on waste disposal at workplace
- GS2.** complete statutory documents relevant to safety and hygiene
- GS3.** fill safety formats for near miss, unsafe condition
- GS4.** identify potential safety risk and report to appropriate authority
- GS5.** communicate and collaborate with others to incorporate sustainable practices
- GS6.** interpret general safety guidelines, labels, charts and signage

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt safe practices at workplace</i>	13	19	-	-
PC1. explain the requirements for safe work area	2	-	-	-
PC2. identify and report any hazards, risks or breaches in site safety to the appropriate authority	2	3	-	-
PC3. follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk	1	2	-	-
PC4. use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition	2	4	-	-
PC5. follow safe practices when working at height and in confined space	1	1	-	-
PC6. handle all required tools, tackles, materials and equipment safely	1	2	-	-
PC7. identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures	1	2	-	-
PC8. apply ergonomic principles wherever required	1	2	-	-
PC9. use safety signs, labels, charts and notices at workplace	1	1	-	-
PC10. identify work safety procedures and instructions for handling heavy components	1	2	-	-
<i>Follow emergencies, rescue and first aid procedures</i>	4	4	-	-
PC11. follow emergency and evacuation procedures in case of accidents, fires and natural calamities	1	1	-	-
PC12. use appropriate fire extinguishers for different types of fire	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.	1	1	-	-
PC14. use correct method to move injured person during an emergency	1	1	-	-
<i>Follow good housekeeping practices and infection control guidelines</i>	4	6	-	-
PC15. follow recommended personal hygiene, workplace hygiene and sanitation practices	1	1	-	-
PC16. clean and disinfect all material, tools and supplies before and after use	1	1	-	-
PC17. report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues	1	2	-	-
PC18. follow processes specified for disposal of hazardous waste	1	2	-	-
NOS Total	21	29	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0106
NOS Name	Maintain Personal Health & Safety at project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health & Safety , Installation, Solar Panel Installation Technician
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council/Awarding Body. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC/AB will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC/AB.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the assessment, every trainee should score the Recommended Pass % aggregate for the Qualification.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0164.Perform Site Survey for installation of Solar PV System for Agrovoltaic system	30	20	-	-	50	13
SGJ/N0165.Procure Solar PV system components according to Agrovoltaic Installation	23	27	-	-	50	13
SGJ/N0166.Introduce tools and tackles used in Solar PV installation for Agrivoltaic system installation	28	22	-	-	50	13
SGJ/N0167.Install Electrical components of Solar PV System for Agrivoltaic system installation	65	35	-	-	100	25
SGJ/N0168.Test and Commission Solar PV System and inspect Agrivoltaic System.	18	32	-	-	50	12
SGJ/N0106.Maintain Personal Health & Safety at project site	21	29	-	-	50	12
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	12
Total	205	195	-	-	400	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.